ECOR2050: Design and Analysis of Engineering Experiments

Assignment 2: Due October 14, 2026

Carleton UNIVERSITY
Canada's Capital University

Instructions:

1. Solutions should be compiled in a single file and submitted to Brightspace.
2. Answer all parts of each question.
3. Show all manual equations, calculations, and units.
4. Plots must be labelled, clear, and legible.
5. Plots must be completed in software.
6. Every solution must include a concluding statement that interprets the numeric result. Correct numeric results without a concluding statement will lose marks.

Q1. A small engineering firm needs to estimate how many computers they need to budget for each year. Based on known failure and replacement rates, the following table was compiled, showing the likelihood of a certain number of computers needing replacement in a given year.

# of computers	Likelihood this # of computers need replacing this year
0	10%
1	15%
2	20%
3	35%
4	20%

(a) What is the probability that the firm will need to replace at least 2 computers next year? **[2 marks]**

(b) How many computers should the company budget for on average? Keep in mind that the answer should be a whole number – you cannot replace a fraction of a computer. Be conservative, making sure that the company's computing needs are met. **[2 marks]**

(c) Find the standard deviation of the annual expected number of computers to replace. **[3 marks]**

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Q2. Given a probability density function defined as follows:

$$f(x) = \begin{cases} 0 & x < -3 \\ a - bx^2 & -3 \le x < 3 \\ 0 & x \ge 3 \end{cases}$$

(a) If the pdf is constrained to be continuous (i.e. $f(x) = 0$ at $x = \pm 3$), show that the values of the constants are $a = 1/4$ and $b = 1/36$. **[3 marks]**

(b) Find the variance of the probability distribution in terms of a and b. **[3 marks]**

(c) What is the probability that a random variable with this distribution has a value lower than -1.0? **[2 marks]**

Q3. The reaction time of a driver to visual stimulus is normally distributed with a mean of 0.30 seconds and variance of 0.0045 s^2.

(a) How likely is it that a driver reacts within less than 0.22 seconds? **[2 marks]**

(b) What percentage of drivers take longer than 0.42 seconds to react? **[2 marks]**

(c) To ensure safety, we want to give sufficient time for most drivers to react. What reaction time is slower than 99.7% of the population? **[2 marks]**